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(57) Abstract :

A method to generate images and a set of segmentation masks corresponding to the generated images and a system thereof. Abstract Disclosed are techniques to generate images and a set of segmentation masks (16) corresponding to the generated images. Techniques include training an LDM and fine tuning said LDM (10) by means of a first low-rank adapter (4) to generate images (9), extracting a set of cross-attention blocks and storing a subset of cross attention blocks in an attention store (11) and generating bounding boxes (8) based on these subsets. Techniques further include filtering the generated images (9) and the bounding boxes (8) corresponding to said generated images(9) by a filtering module (2) and training the segmentation model (3) and fine tuning said segmentation module by a second low-rank adapter (5). The low rank adapters are swappable. Figure 1

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